MUGS-SLAM: Multi-Camera Gaussian Splatting SLAM

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Abstract

We present MUGS-SLAM, a Multi-Camera Gaussian Splatting SLAM system capable of generating highly-accurate 3D reconstructions from sparse views and exploiting its explicit representation and multi-view fisheye setup to enhance state estimation accuracy and robustness in challenging scenarios. Our results demonstrate the feasibility of employing a unified 3D Gaussian Splatting representation for tracking and mapping along multiple cameras, reducing the number of passes required to generate complete maps and maximizing loop closure candidates. In addition, the proposed histogram matching technique enabled the use of high-resolution details in depth priors with negligible overhead, further increasing the reconstruction quality.

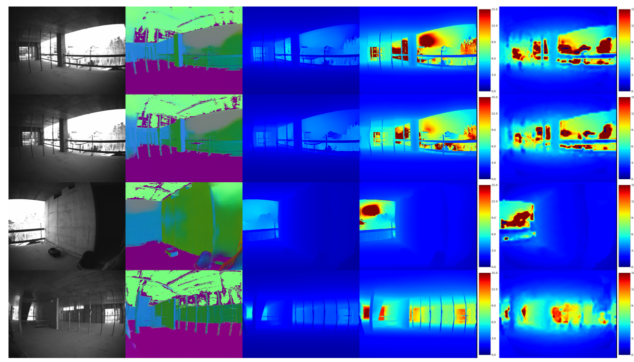

Figure 1. From left to right: RGB inputs, prior normals and depths, histogram-aligned depths, and Bundle-Adjustment depths. To improve the alignment between predicted depths and Gaussian splats, MUGS-SLAM uses the Joint Depth and Scale Alignment (JDSA) algorithm together with histogram matching.

1. Introduction

Simultaneous Localization and Mapping (SLAM) has revolutionized various fields, including robotics, augmented reality, and autonomous navigation, allowing devices to simultaneously estimate their position and build maps of their surroundings [14, 23]. Traditional SLAM approaches often rely on sparse feature-based or semi-dense representations, limiting their ability for photorealistic rendering and accurate dense reconstruction [6]. Recent advances in novel view synthesis, particularly 3D Gaussian Splatting (3DGS) [8], have demonstrated substantial promise in overcoming these limitations, offering memory-efficient continuous representations suitable for real-time applications.

This paper presents MUGS-SLAM, a novel SLAM framework employing Multi-Camera Gaussian Splatting to achieve highly accurate 3D scene reconstructions from sparse camera views. Our system leverages multiple cameras to maximize the field-of-view and improve robustness, especially in challenging conditions such as low-light environments and areas with sparse features [10]. By integrating multi-view inputs with a unified 3D Gaussian Splatting representation, our approach significantly enhances pose estimation accuracy, mapping completeness, and loop closure detection.

Our contributions include a specialized histogram matching technique designed to standardize depth priors across multiple cameras, ensuring robust fusion and consistency in the global Gaussian map. In addition, we validate MUGS-SLAM through comprehensive experiments utilizing custom synthetic datasets generated via the Habitat Simulator [19], designed to replicate realistic indoor scenarios. Our results demonstrate that the proposed multi-camera 3DGS approach effectively combines high-fidelity photorealistic mapping with accurate and reliable pose tracking, outperforming baseline methods and providing a scalable solution for real-time SLAM applications.

The main contributions of this work are summarized as:

1. **Multi-Camera 3DGS SLAM:** We extended single-camera 3DGS approaches to support synchronized multi-camera systems. Our method builds a unified, globally consistent 3D map that avoids cross-view misalignments and supports loop closures across views.
2. **Histogram-Based Depth Alignment:** We introduced a novel histogram matching pipeline integrated with JDSA to enforce per-frame depth scale consistency across different cameras. This approach significantly improved the fusion quality of monocular depth priors.

3. **Synthetic Multi-Camera Dataset:** We adapted Habitat-Sim to simulate multi-camera trajectories in photorealistic 3D environments for benchmarking.

2. Related Works

2.1. 3D Gaussian Splatting SLAM

Monocular and stereo SLAM methods, such as ORB-SLAM2 [14] and LSD-SLAM [6], traditionally focus on sparse or semi-dense reconstructions. Recent developments in neural implicit representations, notably 3D Gaussian Splatting (3DGS), allow highly detailed and photorealistic reconstructions suitable for real-time operations.

Matsuki et al. introduced MonoGS [12], pioneering real-time monocular SLAM with Gaussian splats for live dense reconstruction and direct optimization. Yan et al.'s GS-SLAM [25] further advanced these methods with adaptive Gaussian representation and enhanced speed for pose tracking. Additionally, Sandström et al. developed Splat-SLAM [18], globally optimizing Gaussian representations exclusively from RGB data. However, few works have explored embedding a 3DGS-based explicit representation directly into a SLAM pipeline, highlighting an opportunity to fuse accurate pose tracking with high-fidelity dense mapping.

2.2. Multi-camera SLAM

Exploiting multi-camera systems greatly improves SLAM robustness and increases field-of-view. Frameworks such as MultiCol-SLAM [23] established foundational methodologies for synchronized multi-camera arrangements and generalized monocular frameworks to arbitrary camera rigs by jointly modeling camera intrinsics and extrinsics. Meanwhile, more robust pipelines, such as [11], incorporate wide-angle lenses for increased coverage. Similarly, BAMF-SLAM [30] integrates visual-inertial data and fisheye imagery, significantly enhancing mapping accuracy. However, these solutions often retain sparse representations, contrasting with our dense 3DGS-based framework which enables broad coverage and photorealistic mapping.

2.3. Factor Graphs and Bundle Adjustment

Factor graph optimization and incremental solvers, prominently represented by libraries such as GTSAM [4], iSAM2 [7], and g2o [9], underpin the backend optimization of modern SLAM systems. Our approach specifically adapts these tools to support Gaussian splatting representation efficiently, particularly in multi-camera configurations.

2.4. Photorealistic Mapping

High-fidelity photorealistic mapping has advanced significantly through methods such as Neural Radiance Fields (NeRF) [13] and its follow-ups (e.g. Mip-NeRF [2]), which achieve state-of-the-art novel-view synthesis at the cost of expensive per-scene training. Traditional mesh-based texturing methods trade realism for speed [3], while recent hybrid approaches like RegNeRF [15] attempt to strike a balance between realism and computational efficiency. Gaussian Splatting uniquely enables incremental updates while maintaining photorealistic rendering quality close to NeRF, thus fitting ideally into real-time SLAM pipelines.

3. Methods

Our approach addresses the joint estimation of the continuous trajectory of a multi-camera rig, represented by a sequence of rigid body transformations $\{\mathbf{T}_t\} \subset \mathrm{SE}(3)$, alongside constructing a photorealistic 3D map modeled via 3D Gaussian Splatting (3DGS) [8]. Each pose $\mathbf{T}_t$ comprises rotation and translation components of the camera array at time t, while the map $\mathcal{G} = \{g_i\}_{i=1}^{M}$ contains Gaussian primitives characterized by spatial means $\mu_i \in \mathbb{R}^3$ and covariance matrices $\Sigma_i \in \mathbb{R}^{3\times 3}$.

Using the camera intrinsic matrix $\mathbf{K}$, we denote the perspective projection by $\pi(\cdot; \mathbf{K})$. Each Gaussian splat g_i is projected into image space via:

$$\mathbf{u}_{i,t} = \pi\left(\mathbf{K}\mathbf{T}_t g_i\right),$$

and contributes to image formation through a differentiable blending operation. The observed pixel intensities are assumed to result from this compositing process, allowing optimization of camera poses $\{\mathbf{T}_t\}$ through Sliding-Window Bundle Adjustment (BA) and subsequently of map parameters $\mathcal{G}$. In this approach, MUGS-SLAM represents camera poses and landmarks as nodes in a factor graph. Reprojection constraints, co-visibility edges, and prior-based factors are incrementally added as new keyframes arrive, facilitating efficient incremental updates and local optimizations. In addition, this method enforces long-term loop-closure consistency with minimal overhead, where relative pose constraints between looped keyframes are optimized using a reduced-dimensional Hessian, delivering fast convergence while correcting accumulated drift.

The proposed framework, illustrated in Fig. 2, ingests synchronized streams from N cameras with known intrinsic $\mathbf{K}_n$ and extrinsic pose relative to a common rig frame. Processing is divided into four main components: Keyframe Selector (Sec. 3.1), Front-End (Sec. 3.2), Back-End (Sec. 3.3), and 3DGS-based Mapping (Sec. 3.4).

3.1. Keyframe Selector

Incoming frames are processed at reduced resolution through a compact convolutional network, based on DROID-SLAM [22], to extract salient features and compute selection scores based on feature saliency and dense optical-flow magnitude. Frames surpassing a defined threshold are designated as keyframes, triggering incremental updates

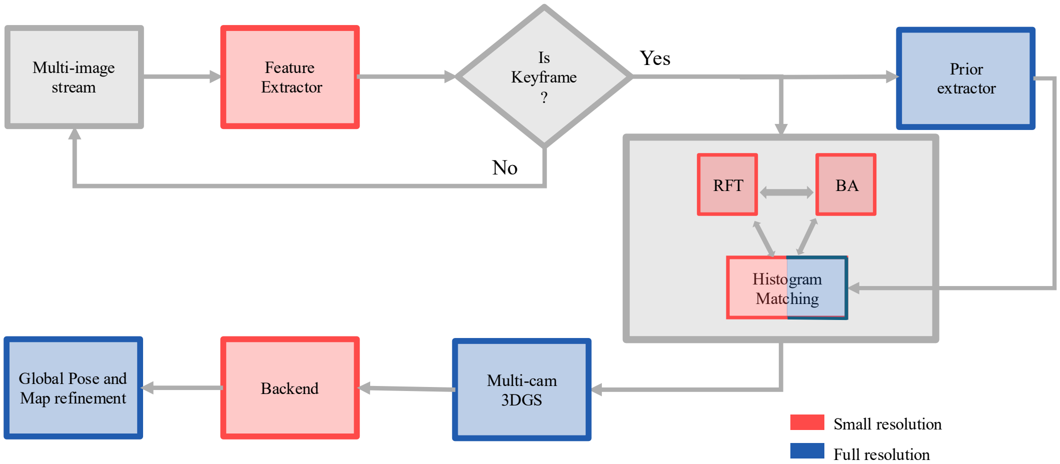

Figure 2. System diagram of MUGS-SLAM. Input frames are preprocessed and sent to a DroidNet model for feature extraction at reduced resolution. A keyframe selector streams the high-confidence frames to a prior extraction module to compute depth and normal maps at full resolution. These keyframes undergo recurrent field transformations (RFT), bundle adjustment (BA), and histogram-based depth alignment to ensure per-frame scale consistency and optimize the estimated pose. The aligned priors are sent to the 3DGS module to generate photorealistic 3D reconstructions. The estimated poses are then iteratively refined to ensure track consistency and map fidelity.

to the factor graph and the computation of full-resolution optical-flow for subsequent alignment.

3.2. Front-End (SLAM)

Operating on recent keyframe pairs, the front-end module computes dense optical-flow and evaluates temporal and spatial similarity (i.e., co-visibility) between frames using a Recurrent Field Transform (RFT) network [21], allowing the system to robustly track features in low-textured and fast-moving scenarios. Based on this, MUGS-SLAM selects high-confidence feature matches to instantiate new factors in the factor graph, annexing the corresponding edges and executing local BA over a sliding window of keyframes. This module continually updates the graph to maintain accurate local pose estimates. It is worth noting that we extend the classical bundle adjustment by jointly optimizing the poses of all N cameras within a sliding window. Cross-view correspondences from available stereo pairs are incorporated as reprojection residuals in a single sparse non-linear least-squares problem, ensuring consistency across viewpoints.

To accelerate computation and achieve real-time throughput, despite the per-pixel formulation at high resolution, MUGS-SLAM employs custom CUDA kernels that perform forward and inverse projections for pinhole and fisheye lenses, as well as undistort functions, directly on the GPU. By leveraging warp-level primitives and minimizing memory transfers, these kernels achieve real-time throughput even at high resolution. Similarly, pairwise covisibility between recently processed keyframes is computed in parallel across all camera streams.

To enhance monocular depth estimation, MUGS-SLAM employs a Joint Depth and Scale Alignment (JDSA) strategy, inspired by [31], that compensates for the inherent scale ambiguity and spatial distortions of monocular priors. Specifically, a spatially-varying 2D scale grid $\mathbf{s}_i \in \mathbb{R}^{m\times n}$ is estimated for each depth prior $\hat{\mathbf{d}}_i$, and the aligned depth is obtained via bilinear interpolation $B_i(\mathbf{p}_i, \mathbf{s}_i)$:

$$r_d = \|\hat{\mathbf{d}}_i \cdot B_i(\mathbf{p}_i, \mathbf{s}_i) - \mathbf{d}_i\|^2, \tag{1}$$

where $\mathbf{d}_i$ is the optimized depth map and $\mathbf{p}_i$ the image coordinates. The optimization objective for JDSA, interleaved with local bundle adjustment, jointly minimizes reprojection and depth prior alignment errors:

$$\begin{aligned} \arg\min_{\mathbf{s},\mathbf{d}} \sum_{(i,j)\in E} & \left\|\check{\mathbf{p}}_{ij} - \Pi\left(\mathbf{T}_{ij}\Pi^{-1}(\mathbf{p}_i, \mathbf{d}_i)\right)\right\|^2_{\Sigma_{ij}} \\ & + \sum_{i\in V} \left\|\check{\mathbf{d}}_i \cdot B_i(\mathbf{p}_i, \mathbf{s}_i) - \mathbf{d}_i\right\|^2, \end{aligned} \tag{2}$$

where $\mathbf{T}_{ij}$ denotes the relative pose between keyframes i and j, and Π is the camera projection function. Efficient solving is enabled via the Schur complement, separating updates to depth $\Delta\mathbf{d}$ and scale $\Delta\mathbf{s}$. This module substantially improves depth alignment and reduces scale drift in monocular SLAM systems.

3.3. Back-End (Global Bundle Adjustment)

A global optimization periodically refines camera poses and map parameters over the entire factor graph, ensuring long-term consistency and correcting drift by tightly coupling trajectory estimates $\{\mathbf{T}_t\}$ to the Gaussian splat map $\mathcal{G}$.

3.4. Mapping (Multi-Camera 3DGS)

MUGS-SLAM employs 3D Gaussian Splatting (3DGS) as its core scene representation, enabling fast rendering and map updates. A unified 3DGS-based map is optimized concurrently with pose refinement, where each Gaussian splat is placed and incrementally updated in the shared rig coordinate frame. This process runs in parallel with the back-end, enabling on-the-fly refinement of geometry and appearance for photorealistic reconstruction.

To ensure global consistency, a single Gaussian splatting map is constructed rather than individual camera-specific maps. This strategy presents two main benefits: (i) prevents cross-view misalignments, as independent maps often exhibit misalignments in overlapping regions that lead to visible seams, whereas a unified Gaussian set $\mathcal{G}$ enforces global spatial coherence, and (ii) it ensures uniform Gaussian density distributions throughout the reconstructed volume, as separate maps may allocate uneven Gaussian densities, resulting in over- or under-smoothed areas. Fusing and pruning allows a uniform density $\forall\, \mathbf{p} \in \mathbb{R}^3$ to be spatially balanced, which is represented as:

$$\sum_{g \in \mathcal{G}_{\text{fused}}} w_g \kappa(\mathbf{p}; \mu_g, \Sigma_g) \tag{3}$$

In multi-camera 3DGS, each depth prior may exhibit a distinct scale and distribution due to sensor characteristics or viewpoint biases. To enforce consistency across views, we perform histogram matching (also known as histogram specification) between the incoming prior and a reference distribution. Let $D_i : \Omega \to \mathbb{R}$ be the continuous depth function at pixel $x \in \Omega$ from camera i, and let $p_i(d)$ denote its probability density function defined as:

$$p_i(d) = \frac{\kappa_d}{mn},$$

where κ_d represents the number of pixels at distance d, and mn is the product of the image dimensions (i.e., the total number of pixels). The corresponding cumulative distribution functions (CDFs) are defined as:

$$F_i(d) \;=\; \int_{-\infty}^{d} p_i(s)\,\mathrm{d}s\,, \qquad F_{\text{ref}}(d) \;=\; \int_{-\infty}^{d} p_{\text{ref}}(s)\,\mathrm{d}s,$$

where p_{ref} is the reference prior obtained from the prior depth distribution. Histogram matching then computes a remapped depth

$$D_i'(x) \;=\; F_{\text{ref}}^{-1}\big(F_i\big(D_i(x)\big)\big),$$

so that the CDF of D_i' aligns exactly with F_{ref}. Aligning these maps is crucial for three reasons:

1. **Scale Consistency**: Without histogram matching, slight differences in depth scaling lead to misaligned Gaussian means μ_j and inconsistent covariances Σ_j, degrading the fidelity of the fused 3DGS map.
2. **Robust Fusion**: By matching distributions, we ensure that the photometric and geometric priors from each camera contribute homogeneously to the global map, avoiding artifacts due to over- or under-represented depth/normal regions.
3. **High-resolution details**: By employing histogram matching, the small details of the high-resolution prior are enhanced with the scale consistency of the BA-refined depth estimate.

The resulting matched priors D_i' are then used to initialize and update the parameters of each Gaussian splat in $\mathcal{G}$, yielding a unified, photorealistic reconstruction. To reduce the impact of any artifacts originating from the fisheye distortion model when computing the CDFs, the current version of MUGS-SLAM undistorts the keyframes and streams the depth-aligned frames to the 3DGS module.

4. Experiments

4.1. Implementation Details

To guarantee reproducible results, facilitate development, and ensure consistent performance across devices, we provide a Docker environment that successfully combines the required software stack with Ubuntu 22.04, Pytorch 2.2.1, CUDA 11.8, CUDNN 8, and Python 3.10. In addition, all the necessary files to reproduce our results are available in our public repository `https://github.com/fgarciacardenas/multi-3dgs`.

The evaluations were performed on a system with an Nvidia RTX 4080S GPU and Intel Core i7-14700KF CPU. For map refinement optimization, we used a depth loss $\lambda_d = 0.1$ for HILTI and Habitat and a downsampling factor $\psi = 64$ for all datasets.

4.2. Datasets

To evaluate the performance of MUGS-SLAM in a controlled, photorealistic environment, we generated a custom synthetic dataset using Habitat-Sim [16]. We configured the simulator to mimic the geometry and appearance of indoor rooms, selecting a diverse set of layouts (e.g., furnished living rooms, narrow corridors, and open-plan offices). Within each scene, we instantiated our virtual multi-camera rig composed of N synchronized sensors with a resolution of 800×600 px. During data collection, the rig followed predetermined trajectories that included straight traversals, on-spot rotations, and loops. For each keyframe, we recorded RGB images alongside ground-truth (GT) poses at 15 Hz.

To compare the system performance in real-life settings, we chose Hilti-22 [27] which uses four grayscale fisheye cameras in construction sites and deals with drastic changes in exposure and scene depth. It is worth noting here that there is a lack of publicly available multi-camera datasets which contain GT data for adequate evaluation.

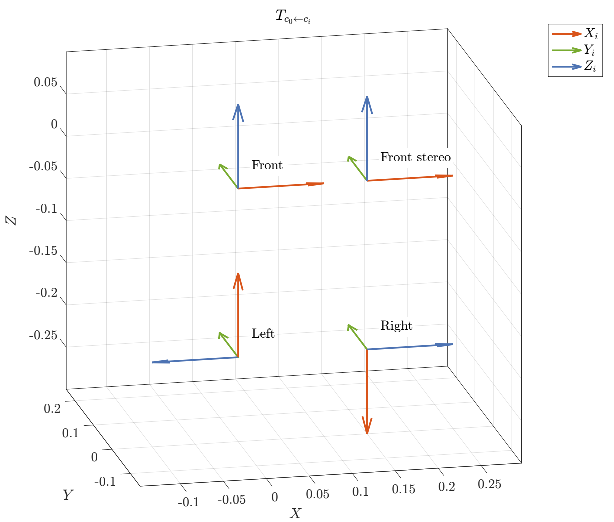

Figure 3. Visualization of the extrinsics of the configuration chosen for the simulated datasets.

4.3. Evaluation Metrics

We assess the performance of our system along two dimensions: camera tracking accuracy and appearance fidelity. To evaluate the accuracy of estimated camera trajectories, we use the *Absolute Pose Error* (APE), which measures the root mean square error (RMSE) between estimated and ground truth camera poses after alignment. This metric reflects the global consistency of the estimated trajectory and is widely used in SLAM benchmarks [32].

To evaluate photometric fidelity of rendered keyframes, we use three established image similarity metrics: (i) PSNR (Peak Signal-to-Noise Ratio), which measures the logarithmic ratio between the maximum possible pixel value and the mean squared error (MSE) between the rendered and ground truth images, (ii) SSIM (Structural Similarity Index), which captures structural similarity between images by comparing luminance, contrast, and structure [24], and (iii) LPIPS (Learned Perceptual Image Patch Similarity), a deep-learning-based metric that correlates well with human perceptual judgments, computed using features from pre-trained neural networks [28].

4.4. Tracking and Reconstruction Performance

In this section, we analyze the tracking and reconstruction performance of MUGS-SLAM on three datasets (Habitat room09, room14, and Hilti exp04), comparing it quantitatively and qualitatively against state-of-the-art SLAM systems HI-SLAM2 and BAMF-SLAM.

Table 1 summarizes the Absolute Pose Error (APE) results, demonstrating superior tracking performance by MUGS-SLAM across all tested scenarios. This improvement primarily arises from our multi-camera BA framework, which robustly handles sparse features and complex scene geometry. Similarly, Fig. 4 depicts the aligned pose error for Hilti Exp04, reinforcing the tracking accuracy improvements and the consistency throughout the trajectory.

Table 1. Absolute Pose Error (APE)

Dataset	HI-SLAM2	BAMF	MUGS
room09	2.655	0.475	0.370
room14	0.320	0.221	0.077
exp04	0.847	0.177	0.126

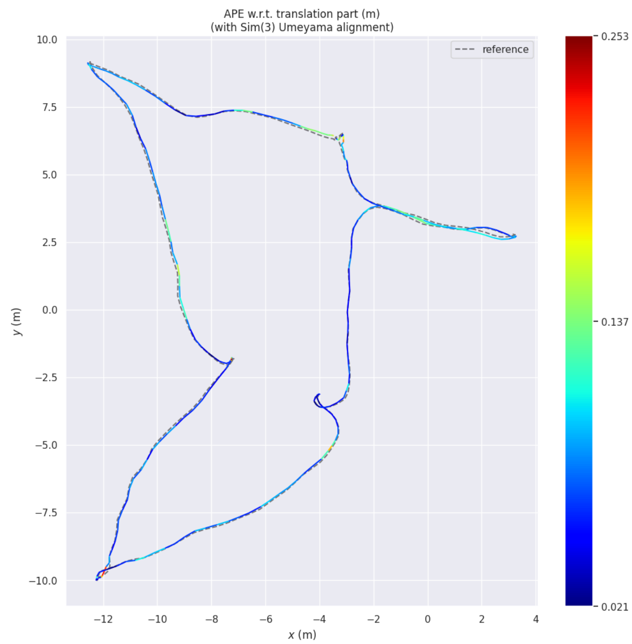

Figure 4. Aligned pose error in Hilti Exp04, where MUGS-SLAM achieved accurate tracking throughout the dataset.

Table 2 compares reconstruction quality using standard photometric metrics. While MUGS-SLAM generally matches HI-SLAM2 on SSIM and LPIPS, we observed lower PSNR scores on the Hilti sequence. This lower PSNR score is attributed to conflicting optimization constraints arising from multiple camera views, affecting per-pixel reconstruction accuracy, particularly in challenging grayscale environments. In this regard, it is worth noting that PSNR alone may not be a fair metric to compare these algorithms since MUGS-SLAM is able to reconstruct more of the scene thanks to its side cameras, which can lead to conflicting losses between the views and reduce the overall PSNR.

Qualitative results further validate the strength of MUGS-SLAM. Fig. 5 presents the 3DGS reconstructions of the rooms 09 and 14, respectively, showcasing the detailed and coherent structures achieved by the proposed method.

Table 2. Reconstruction Metrics

Method	**Metric**	room09	room14	exp04
HI-SLAM2	PSNR[dB]↑	28.71	34.56	26.07
	SSIM↑	0.867	0.955	0.859
	LPIPS↓	0.357	0.117	0.499
MUGS-SLAM	PSNR[dB]↑	30.97	32.6	21.80
	SSIM↑	0.886	0.944	0.827
	LPIPS↓	0.272	0.163	0.560

(a) Reconstruction of Room 09 using MUGS-SLAM.

(b) Reconstruction of Room 14 using MUGS-SLAM

Figure 5. 3DGS-based reconstructions of Rooms 09 (a) and 14 (b) of the synthetic Habitat dataset using MUGS-SLAM.

Finally, Fig. 6 clearly demonstrates MUGS-SLAM's superior reconstruction quality compared to HI-SLAM2, with sharper and more detailed reconstructions, particularly evident in scene textures and edges. This behavior is due to our histogram matching technique, which allows the MUGS-SLAM system to capture finer details.

5. Work Distribution

The work was divided among the authors as follows:

- **Facundo Garcia**: Prepared Docker containers, improved 3DGS methods for multi-camera, reworked TSDF, implemented data loaders, executed 3DGS benchmarks, and focused on improving overall stability.
- **Luca Monegaglia**: Worked on BA and JDSA algorithms, executed pose estimate benchmarks, worked on feature and prior extractors.

(a) MUGS-SLAM reconstruction on Room14

(b) HI-SLAM2 reconstruction on Room14

Figure 6. Comparison of the reconstruction quality between MUGS-SLAM (a) and HI-SLAM2 (b) on the Room 14 dataset.

- **Alessandro Petitti**: Generated the synthetic datasets, worked on 3DGS, and assisted with benchmarking.
- **Elia Raimondi**: Worked on multi-camera factor graph, improved BA and JDSA algorithms, implemented histogram matching, optimized CUDA kernels.

6. Conclusions

We introduced MUGS-SLAM, a novel multi-camera SLAM framework based on 3D Gaussian Splatting, capable of producing photorealistic and geometrically consistent 3D reconstructions. By unifying multi-view inputs within a single global Gaussian map, MUGS-SLAM achieves robust camera tracking and high-fidelity scene representation in both synthetic and real-world environments.

Key innovations include a histogram-based depth alignment strategy to standardize priors across viewpoints, a GPU-accelerated bundle adjustment backend supporting both fisheye and pinhole models, and a custom synthetic dataset pipeline using Habitat-Sim for multi-camera simulation and evaluation. Experimental results across multiple datasets validate the system's accuracy in trajectory estimation and its capacity for rendering high-quality maps.

7. Future Work

Building upon the foundations laid by MUGS-SLAM, future research will explore supporting heterogeneous sensor suites, real-time 3DGS mapping, robustness to outliers and dynamic scenes, and advanced neural prior estimators.

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